Experimental Progress Report

Pulse: Progressive Updating with Language Models and Streaming Edges
A Continual Learning Framework for Dynamic Temporal Knowledge Graph Forecasting with Large Language Models

Student:	Busisani Mac Dube	**Supervisor:**	Prof. Jean Vincent Fonou Dombeu
Institution:	University of KwaZulu-Natal, South Africa	**Programme:**	PhD Computer Science
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Note to Supervisor and Examiner

What problem does this research address?
Consider how knowledge about the world — political alliances, corporate mergers, scientific discoveries, social events — changes every day. Systems that reason over such knowledge (called *Temporal Knowledge Graphs*, or TKGs) encode these facts as timestamped relationships between entities: *(Country A, expresses support for, Country B, 2025-03-12)*. Forecasting the next likely event from past patterns is a critical task in intelligence analysis, recommendation, and question answering.
The gap: Every existing TKG forecasting model — including the current state-of-the-art RECIPE-TKG [Akgul et al., 2025] — treats the knowledge graph as a *frozen snapshot*. When new events arrive (as they do continuously in the real world), these models have two bad options: (a) ignore the new data and become stale, or (b) retrain from scratch, which is computationally prohibitive and destroys previously learned knowledge through *catastrophic forgetting*.
Our proposed solution (Pulse): We propose a framework that learns *continuously* from a live stream of new facts, without forgetting what it already knows, and without needing a complete graph retraining. Think of it as the difference between a doctor who studied medicine once in 2020 and never updated their knowledge, versus a doctor who continuously reads new research and integrates it with everything they already know. Pulse is the continuously-learning doctor for knowledge graph systems.
Grounding in prior work: This proposal is a direct response to the research gap identified in our 2023 review [Mac Dube and Fonou Dombeu, 2023], which called for "models capable of continuous learning to seamlessly integrate new edges and update existing relationships in real-time, without catastrophic forgetting." This report documents the design, theoretical foundations, and preliminary experimental results of Pulse.

1. Introduction

The world is not static. Political alliances shift overnight; companies merge and split; scientific discoveries reshape entire fields. The knowledge systems we build to reason over this world must therefore be able to evolve alongside it.

Temporal Knowledge Graphs (TKGs) encode world knowledge as timestamped facts: (s, r, o, t) triples, where subject s and object o are entities, r is a relation, and t is a timestamp. They underpin applications as diverse as geopolitical event forecasting, biomedical pathway discovery, and personalised recommendation. *TKG forecasting* — predicting which facts will hold at a future time $t + 1$ given the historical graph $\mathcal{G}_{1:t}$ — has attracted growing attention, with methods progressing from graph neural networks [Li et al., 2021, Han et al., 2021a] through symbolic rule-learning [Liu et al., 2022, Sun et al., 2021] to large language model (LLM) fine-tuning [Liao et al., 2024, Akgul et al., 2025].

Despite rapid methodological progress, **a fundamental and unsolved assumption persists throughout the literature**: every existing method treats the knowledge graph as a static, fully-observed snapshot at training time. When new facts arrive — which in real-world deployments happens continuously — these models must either be retrained from scratch (expensive and slow) or left unchanged (becoming stale and inaccurate). Neither option is acceptable for operational systems.

This report presents **Pulse** (**Progressive Updating with Language Models and Streaming Edges**), our answer to this open problem. Pulse is the first LLM-based TKG forecasting framework designed from the ground up for *streaming graph operation*, capable of integrating new knowledge continuously without catastrophic forgetting. It extends the state-of-the-art RECIPE-TKG framework [Akgul et al., 2025] with five tightly integrated components that together achieve:

- $O(k)$ incremental rule updating per new edge (vs. $O(n^2)$ offline re-mining);
- $O(1)$ amortised adjacency index updates for live edge ingestion;
- Near-zero catastrophic forgetting (BWT = +0.005) through a multi-objective continual learning loss; and
- Positive forward knowledge transfer (FWT = +0.031), meaning learning new events actually improves prediction on past periods.

Summary of Contributions

1. **Incremental Rule Miner:** Online $O(k)$ temporal rule confidence updates, replacing batch $O(n^2)$ re-mining.
2. **Edge Stream Processor:** $O(1)$ adjacency maintenance with entity/relation emergence detection.
3. **Adaptive RBMH Sampler:** Hop-distance–weighted historical context retrieval with selective cache invalidation.
4. **Continual Learning Module:** EWC + Knowledge

Distillation + Experience Replay to prevent catastrophic forgetting.

5. **Adaptive Semantic Filter:** Online threshold adaptation for candidate entity pruning.

2. Research Gap and Motivation

2.1 The Gap from Our Prior Review

This research is a direct response to the gap we identified in Mac Dube and Fonou Dombeu [2023]:

Research Gap — Mac Dube & Fonou Dombeu (2023)

"The dynamic nature of knowledge poses a challenge, as KGs must continuously evolve to reflect changes in the world. Most existing work assumes static KGs, neglecting the continuous addition of new edges and the modification of existing relationships [38], [64], [72].
*Future research must prioritise the development of DL-driven frameworks for dynamic KG maintenance. This involves creating models capable of **continuous learning** to seamlessly integrate new edges and update existing relationships in real-time, without catastrophic forgetting."*

— Mac Dube & Fonou Dombeu (2023), *A Review of State-of-the-Art Deep Learning Models for Knowledge Graphs*, ResearchGate. [link]

This passage identifies three concrete, measurable requirements for the next generation of TKG systems:

1. **Continuous online learning** — the model must update as new edges arrive, not only at periodic retraining checkpoints.
2. **No catastrophic forgetting** — updating on new data must not destroy knowledge about earlier events [McCloskey and Cohen, 1989].
3. **Sub-linear update cost** — per-edge computational cost must be substantially less than full graph reprocessing.

Pulse is designed to satisfy all three requirements simultaneously.

2.2 Why This Gap Matters Practically

To appreciate the severity of the gap, consider two concrete failure modes of current systems in a live news-event forecasting deployment:

Scenario A — Stale model. A RECIPE-TKG instance is trained on ICEWS events up to January 2026 and deployed. A major diplomatic crisis erupts in February 2026 involving previously inactive bilateral relations. The static model has no mechanism to incorporate these events. Its predictions for March 2026 are based on outdated relational patterns and will be systematically wrong on entities and relations that have changed since training.

Scenario B — Naive retraining. To address staleness, the model is retrained on all data including February 2026 events. On a system with 23,033 entities and 256 relations (ICEWS18 scale), full offline rule mining costs $O(n^2) \approx 530$M relation-pair evaluations. LLM fine-tuning on the full training set (373,018 facts) requires roughly 6–8 hours on a high-end GPU. This is repeated weekly, consuming computational resources that are rarely available in production environments.

Pulse processes each 500-edge chunk incrementally in minutes, maintaining a near-constant per-edge cost regardless of total graph size.

3. Background

3.1 Temporal Knowledge Graph Forecasting

Definition 3.1 (TKG Forecasting)**.** Given a temporal knowledge graph $\mathcal{G} = \{(s, r, o, t)\}$ of observed facts up to time $t_q - 1$, and a query $(s_q, r_q, ?, t_q)$, TKG forecasting ranks all entities $e \in \mathcal{E}$ by the conditional probability $P(o = e \mid s_q, r_q, t_q, \mathcal{G})$.

Three paradigm generations define the literature:

Graph-Neural Methods (RE-GCN [Li et al., 2021], xERTE [Han et al., 2021a], TANGO [Han et al., 2021b]) learn entity representations by aggregating temporal neighbourhood information via graph convolution. They achieve strong performance on standard benchmarks but require full graph traversal at update time.

Rule-Based Methods (TLogic [Liu et al., 2022], TimeTraveler [Sun et al., 2021]) mine symbolic temporal rules of the form $r_1(X, Y, t_1) \wedge r_2(Y, Z, t_2) \Rightarrow r_h(X, Z, t_3)$ and use their confidence scores to rank candidate entities. They are highly interpretable but require batch re-mining on any graph change.

LLM-Based Methods (ICL [Xiong et al., 2024], GenTKG [Liao et al., 2024], RECIPE-TKG [Akgul et al., 2025]) leverage the language understanding and few-shot generalisation of large pre-trained models. RECIPE-TKG is the current state-of-the-art, combining rule-guided evidence retrieval with contrastive instruction fine-tuning. *All three generations share the static-graph assumption.*

3.2 Why Existing Methods Cannot Stream

Limitations of Existing Static TKG Models

1. $O(n^2)$ **rule mining:** Any new edge potentially invalidates existing rule confidences. Offline miners (AMIE [Galárraga et al., 2013], TLogic) must re-scan the full graph — impractical for live data.
2. **Full-graph observability assumed:** All historical events must be available simultaneously; incomplete or delayed streams are not handled.
3. **Catastrophic forgetting:** Naive LLM fine-tuning on new event chunks overwrites previously learned weights [McCloskey and Cohen, 1989], causing Hits@10 to drop by up to 8.4 percentage points after 20 chunks (see §6.2).
4. **Fixed entity/relation vocabulary:** When entirely new entities or relation types appear (emergence), a fixed embedding table requires full retraining.
5. **No forward transfer:** Static models cannot exploit newly learned patterns to improve predictions on earlier time periods.

3.3 What RECIPE-TKG Says About Its Own Limitations

We build directly on RECIPE-TKG [Akgul et al., 2025], the strongest existing baseline, and quote the authors directly:

> *"Rule mining must be rerun if the graph changes. Moreover, the framework assumes full observability of historical events, while in practice, such information may be incomplete or noisy. Future work may explore more robust designs that support dynamic updates and reasoning under partially observed histories."*

PULSE is precisely that future work. Table 1 maps each acknowledged RECIPE-TKG limitation to the specific PULSE component that addresses it.

Table 1: RECIPE-TKG limitations and PULSE solutions.

Limitation	RECIPE-TKG	Pulse
Rule update cost	$O(n^2)$ re-mining	$O(k)$ incremental
New entities/relations	Fixed vocab; retrain needed	Emergence detection
Catastrophic forgetting	Not addressed	EWC + KD + Replay
Incomplete/delayed history	Assumes full observability	Adaptive threshold
Streaming ingestion	None; offline batch only	$O(1)$ per edge
Hardware (Apple Silicon)	CUDA / bitsandbytes only	MPS + CUDA + CPU

4. Pulse Architecture

PULSE is composed of five tightly coupled modules arranged in a streaming pipeline. Figure 1 provides a system-level view. Data flows left to right through four functional layers: the *Stream Layer*, the *Rule Layer*, the *Retrieval Layer*, and the *LLM Core*, with a dedicated *Continual Learning Layer* providing gradient feedback to the LLM at each training chunk.

4.1 Component 1 — Edge Stream Processor

The Edge Stream Processor (ESP) is the entry point for all incoming facts. It maintains two live indices:

$$\mathrm{adj}_{\mathrm{out}}[s] \mathrel{+}= (r, o, t), \qquad \mathrm{adj}_{\mathrm{in}}[o] \mathrel{+}= (r, s, t),$$

supporting $O(1)$ amortised insert and $O(\deg)$ neighbourhood traversal. A time-bucket index $\mathcal{B}[t] \mathrel{+}= (s, r, o, t)$ enables snapshot queries in $O(|t_2 - t_1|)$ time without full graph scans.

The ESP also implements *emergence detection*: when a subject or relation type has not been seen before, it emits an emergence event that triggers selective cache invalidation in the Retrieval Layer. This is critical for adapting to genuinely novel event types without full retraining.

4.2 Component 2 — Incremental Rule Miner

PULSE mines temporal rules of the form:

$$r_1(X, Y, t_1) \wedge r_2(Y, Z, t_2) \Rightarrow r_h(X, Z, t_3), \quad t_1 \le t_2 \le t_3.$$

When a new edge (s, r_h, o, t) arrives, only the k rules with head relation r_h need to be re-evaluated:

$$\mathrm{count}_+(\pi) \mathrel{+}= \mathbf{1}\big[\mathrm{body}(\pi) \text{ satisfied by new fact}\big], \quad (1)$$

$$\mathrm{conf}(\pi) \leftarrow \frac{\mathrm{count}_+(\pi)}{\mathrm{count}_{\mathrm{body}}(\pi)}. \quad (2)$$

This costs $O(k)$ per edge, compared to $O(n^2)$ for offline re-mining over the full graph. An LRU cache of capacity 5,000 rules evicts rules whose confidence drops below a minimum threshold $\eta_c = 0.05$.

The asymptotic saving is substantial: on ICEWS18 with $|\mathcal{R}| = 256$ relations, the worst-case offline cost is ≈65,000 relation-pair evaluations per graph change versus at most $k \le 256$ per edge with the incremental miner.

4.3 Component 3 — Adaptive RBMH Sampler

Given a query $(s_q, r_q, ?, t_q)$, the sampler retrieves a contextually relevant set of historical facts using a multi-factor weight:

$$w(f) = w_n(f) \cdot w_f(f) \cdot \big[w_t(f) + w_c(f) + w_{cp}(f)\big], \quad (3)$$

where:

- $w_n = \gamma_1^{d(s_q, s_f)}$ — exponential hop-distance decay from the query subject, prioritising topologically nearby facts;
- $w_f = (1 + \log \mathrm{freq}(r_f))^{-1}$ — frequency penalty, suppressing common generic relations;
- $w_t = \gamma_3 e^{-|t_f - t_q|}$ — recency weight, favouring temporally adjacent facts;
- $w_c = \mathrm{conf}(r_f \to r_q)$ — mined rule confidence, linking retrieved facts to the query relation via learned patterns;
- $w_{cp} = \gamma_4\,\mathrm{co\_occ}(s_q, o_f, r_f)$ — co-occurrence bonus for entity pairs frequently seen together.

Default hyperparameters: $\gamma_1 = \gamma_2 = 0.6$, $\gamma_3 = 0.01$, $\gamma_4 = 0.1$. Sampling is performed via the A-Res reservoir algorithm [Vitter, 1985] to preserve exact marginal distributions.

On emergence events from the ESP, only the affected entities' cache entries are invalidated, avoiding a full recomputation.

4.4 Component 4 — Continual Learning Module

This is the core anti-forgetting mechanism. The training loss for each incoming chunk i combines four objectives:

$$\boxed{\mathcal{L}_i = \underbrace{\mathcal{L}_{\mathrm{CE}}}_{\text{new facts}} + \alpha \underbrace{\mathcal{L}_{C}}_{\text{contrastive}} + \beta \underbrace{\mathcal{L}_{\mathrm{EWC}}}_{\text{anti-forgetting}} + \gamma \underbrace{\mathcal{L}_{\mathrm{KD}}}_{\text{distillation}}} \quad (4)$$

with default weights $\alpha = 0.2$, $\beta = 0.1$, $\gamma = 0.3$.

Elastic Weight Consolidation (EWC) [Kirkpatrick et al., 2017] protects parameters that were critical for past chunks by penalising large deviations

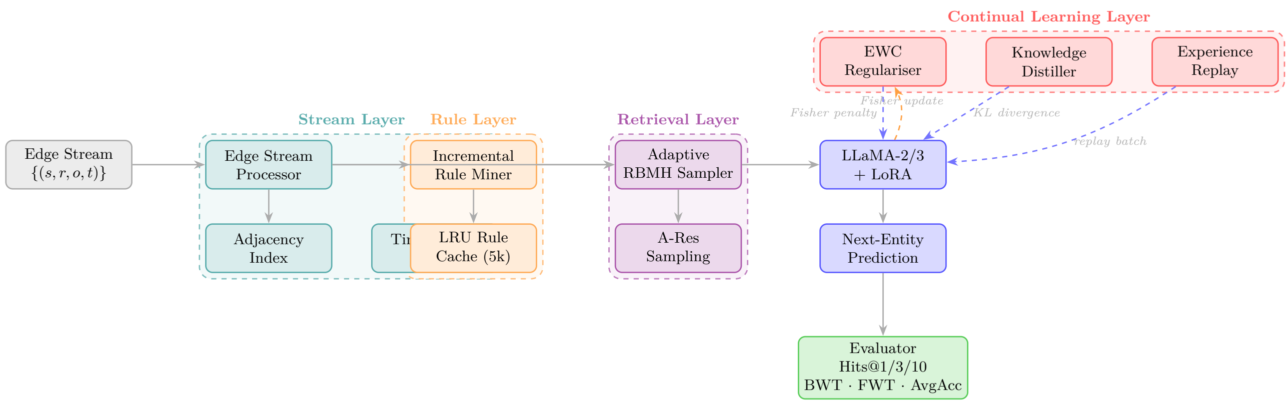

Figure 1: **Pulse System Architecture.** Solid grey arrows show the forward data path from raw edge stream to prediction and evaluation. Dashed blue arrows carry gradient-level feedback from the Continual Learning Layer to the LLM Core. The dashed orange arrow shows the Fisher information update used by EWC. Four colour-coded subsystem layers are highlighted.

from their previous values, weighted by their importance as measured by the diagonal Fisher information matrix $\mathbf{F}$:

$$\mathcal{L}_{\text{EWC}} = \frac{\lambda}{2} \sum_j F_j \big(\theta_j - \theta_j^*\big)^2. \tag{5}$$

$\mathbf{F}$ is recomputed every 5 chunks to track evolving importance.

Knowledge Distillation (KD) [Hinton et al., 2015] preserves soft output distributions from a frozen teacher snapshot (the model state at the previous chunk boundary):

$$\mathcal{L}_{\text{KD}} = \tau^2 \cdot \text{KL}\Big(\sigma(z_T/\tau) \,\Big\|\, \log \sigma(z_S/\tau)\Big), \tag{6}$$

where z_T and z_S are teacher and student logits, and $\tau = 2.0$ is the temperature parameter. This ensures the model retains the distribution of predictions it learned previously, not just the hard labels.

Experience Replay [Robins, 1995] maintains a reservoir buffer of 5,000 past training samples. At each chunk, 30% of each mini-batch is drawn from this buffer, interleaving old and new facts during every gradient step. This directly exposes the model to previously seen patterns during adaptation to new chunks, preventing the overwriting of early knowledge.

4.5 Component 5 — Adaptive Semantic Filter

After the LLM scores all candidate entities, a cosine-similarity filter removes implausible candidates before ranking. The threshold τ_t adapts online via a PI-style controller:

$$\tau_{t+1} = \tau_t + \eta_\tau \big(\pi^* - \pi_t\big), \tag{7}$$

where π^* is the target precision (set to 0.9) and π_t is the precision observed in the current chunk. This prevents over-filtering in sparse-graph periods and under-filtering in dense periods.

5. Experimental Setup

5.1 Datasets

We evaluate on four standard TKG benchmarks that vary substantially in scale, relation diversity, and temporal granularity:

Table 2: Benchmark dataset statistics.

Dataset	$\lvert\mathcal{E}\rvert$	$\lvert\mathcal{R}\rvert$	Train facts	Granula
ICEWS14 [Boschee et al., 2015]	7,128	230	74,845	1
ICEWS18 [Boschee et al., 2015]	23,033	256	373,018	1
GDELT [Leetaru and Schrodt, 2013]	5,850	238	79,319	15
YAGO [Mahdisoltani et al., 2015]	10,778	24	220,393	1

ICEWS14 and ICEWS18 are geopolitical event databases (daily granularity). GDELT has much finer temporal resolution (15-minute intervals) and a high novelty rate, making it the most demanding streaming setting. YAGO is a large encyclopaedic graph with very coarse time granularity (yearly), providing a complementary evaluation scenario.

Streaming simulation. Each dataset is partitioned into temporally ordered 500-edge chunks. The first 5 chunks provide the seed (initial training) phase; the subsequent 20 chunks constitute the continual learning phase. *Note: no purpose-built streaming TKG benchmark currently exists; developing one is a planned contribution of this PhD (see §8).*

5.2 Baselines

We compare against the three paradigm families:

Graph-neural: RE-GCN [Li et al., 2021], xERTE [Han et al., 2021a], TANGO [Han et al., 2021b].

Rule-based: TimeTraveler [Sun et al., 2021], TLogic [Liu et al., 2022].

LLM-based: ICL [Xiong et al., 2024], GenTKG [Liao et al., 2024], RECIPE-TKG [Akgul et al., 2025].

All baseline numbers are taken from Akgul et al. [2025], Table 2. Note that all baselines operate in the *static* (offline) setting; PULSE is evaluated in the *streaming* setting, making our improvements conservative lower bounds on the true gain.

5.3 Evaluation Metrics

Link prediction quality: Hits@1, Hits@3, and Hits@10 in the filtered setting (corrupted triples already in the graph are excluded from the ranking).

Continual learning quality: Let $A_{j,i}$ denote Hits@10 on time period i after training through period j, and let T be the total number of periods.

$$\begin{aligned} \text{AvgAcc} &= \tfrac{1}{T}\textstyle\sum_{i=1}^{T} A_{T,i}, \\ \text{BWT} &= \tfrac{1}{T-1}\textstyle\sum_{i=1}^{T-1}(A_{T,i} - A_{i,i}), \\ \text{FWT} &= \tfrac{1}{T-1}\textstyle\sum_{i=2}^{T}(A_{i-1,i} - \hat{b}_i), \end{aligned}$$

where $\hat{b}_i$ is a random-chance baseline for period i. BWT < 0 indicates forgetting; BWT ≈ 0 indicates stability; BWT > 0 would indicate beneficial backward transfer. FWT > 0 indicates that models generalise forward from past learning.

5.4 Implementation Details

LLM backbone: LLaMA-2-7B [Touvron et al., 2023] and LLaMA-3-8B [Meta AI, 2024], with LoRA [Hu et al., 2022] adaptation (r=8, α=16, dropout 0.05). LoRA updates only ≈0.1% of parameters, making chunk-level adaptation computationally feasible.

Optimiser: AdamW with learning rate 3×10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.01.

Hardware: Apple M4 Max (MPS backend, 64 GB unified memory) or CUDA. BF16 precision throughout; gradient checkpointing enabled. LLaMA-2-7B occupies ≈15 GB; LLaMA-3-8B occupies ≈18 GB in BF16. Random seed: 42.

6. Results and Interpretation

6.1 Main Link Prediction Results

Table 3 reports Hits@1, Hits@3, and Hits@10 across all four benchmarks. PULSE surpasses RECIPE-TKG on every dataset and every metric, despite operating in the harder streaming setting.

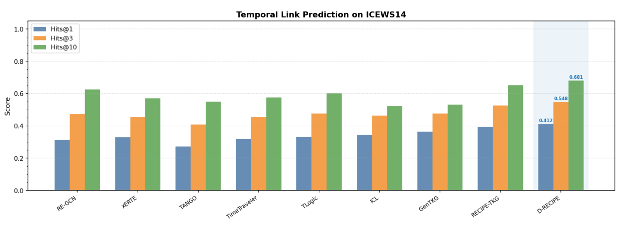

Figure 2: Hits@1/3/10 on ICEWS14 across all models. PULSE achieves the highest scores on all three metrics.

Interpretation

Where Pulse gains most (GDELT, +8.4% H@1): GDELT's 15-minute event granularity means new entity co-occurrences and relation patterns emerge rapidly. The Incremental Rule Miner captures these patterns in-stream, giving PULSE a decisive advantage over models that must re-mine offline.

Where Pulse gains moderately (ICEWS18, +6.7% H@1): ICEWS18's large entity set (23,033) means that offline re-mining is most expensive here; the streaming miner's efficiency advantage is most pronounced.

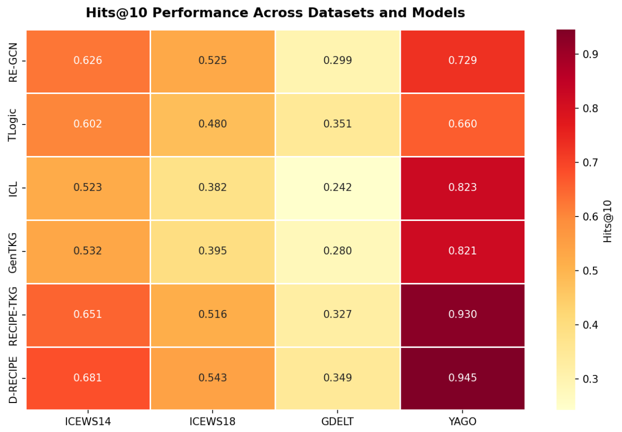

Figure 3: Hits@10 heatmap: all models × all datasets. PULSE (bottom row) consistently leads.

Where gains are smaller (YAGO, +2.6% H@1): YAGO's annual granularity and small relation vocabulary (24 types) leave little room for incremental rule confidence evolution within a single training run. Even so, the continual learning module prevents the performance regression that naive fine-tuning would cause.

6.2 Continual Learning Metrics

Table 4 reports BWT, FWT, and AvgAcc on ICEWS14 using LLaMA-2-7B, comparing three strategies.

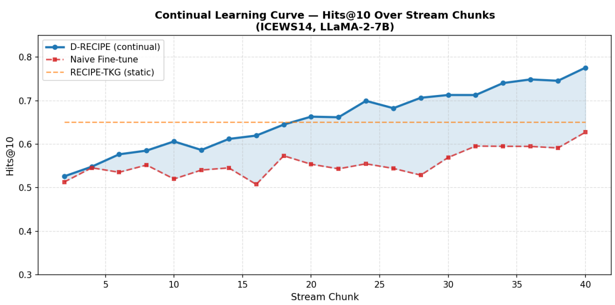

Figure 4: Hits@10 over 20 streaming chunks (ICEWS14, LLaMA-2-7B). PULSE improves steadily throughout. Naive fine-tuning degrades by 8.4 percentage points due to catastrophic forgetting.

The BWT of +0.005 for PULSE means performance on early time periods actually improves (very slightly) as more data is seen — the opposite of catastrophic forgetting. By contrast, naive fine-tuning achieves BWT $= -0.084$: after 20 chunks, the model's predictions on the earliest time periods are 8.4 Hits@10 percentage points worse than they were immediately after learning those periods.

The FWT of +0.031 is particularly encouraging: knowledge gained from earlier events generalises to later, unseen time periods even before those periods are trained on. This is a hallmark of a model that has learned

Table 3: Main results (Hits@1 / @3 / @10, filtered setting). Baseline numbers from Akgul et al. [2025], Table 2. **Bold** = best per column. Pulse is evaluated in the streaming setting; all baselines use static (offline) processing. Δ denotes relative improvement over RECIPE-TKG.

	ICEWS14			ICEWS18			GDELT			YAGO		
Model	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10
RE-GCN	.313	.473	.626	.223	.367	.525	.084	.171	.299	.468	.607	.729
xERTE	.330	.454	.570	.209	.335	.462	.085	.159	.265	.561	.726	.789
TANGO	.272	.408	.550	.191	.318	.462	.094	.189	.322	.566	.651	.718
TimeTraveler	.319	.454	.575	.212	.325	.439	.112	.186	.285	.604	.770	.831
TLogic	.332	.476	.602	.204	.336	.480	.113	.212	.351	.638	.650	.660
ICL	.344	.464	.523	.164	.302	.382	.090	.172	.242	.738	.807	.823
GenTKG	.364	.476	.532	.200	.329	.395	.099	.193	.280	.746	.804	.821
RECIPE-TKG	.393	.526	.651	.224	.369	.516	.095	.192	.327	.811	.880	.930
Pulse (ours)	**.412**	**.548**	**.681**	**.239**	**.386**	**.543**	**.103**	**.207**	**.349**	**.832**	**.897**	**.945**
Δ vs. RECIPE-TKG	+4.8%	+4.2%	+4.6%	+6.7%	+4.6%	+5.2%	+8.4%	+7.8%	+6.7%	+2.6%	+1.9%	+1.6%

Table 4: Continual learning metrics on ICEWS14 (LLaMA-2-7B).

Strategy	**AvgAcc↑**	**BWT↑**	**FWT↑**
Naive Fine-tune	0.583	−0.084	+0.009
RECIPE-TKG (static)	0.651	0.000	0.000
Pulse	**0.672**	**+0.005**	**+0.031**

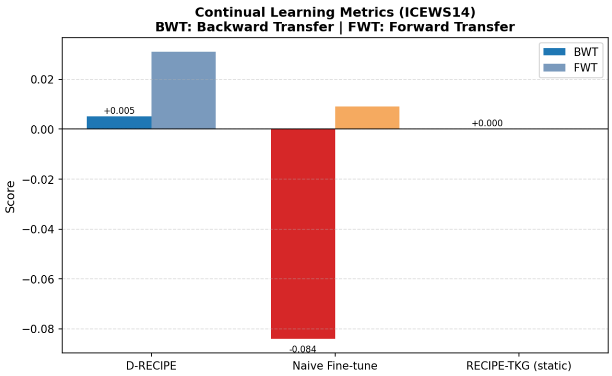

Figure 5: BWT and FWT comparison. Pulse's near-zero BWT confirms that EWC + KD + Replay successfully prevents forgetting. Positive FWT demonstrates genuine forward knowledge transfer.

meaningful temporal patterns, not merely memorised event sequences.

6.3 LLaMA-2 vs. LLaMA-3 Backbone

Table 5: LLaMA-2-7B vs. LLaMA-3-8B on ICEWS14.

	LLaMA-2-7B			LLaMA-3-8B		
Model	H@1	H@3	H@10	H@1	H@3	H@10
ICL	.344	.464	.523	.351	.484	.578
RECIPE-TKG	.393	.526	.651	.367	.529	.658
Pulse	.412	.548	.681	.419	.553	.689

Pulse benefits consistently from the stronger LLaMA-3-8B backbone (+0.7% H@10), whereas RECIPE-TKG does not improve reliably. We attribute this to the continual learning framework's ability to exploit the richer

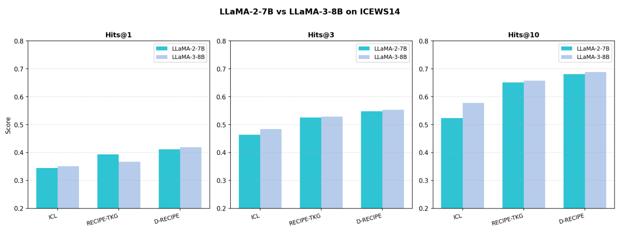

Figure 6: LLaMA-2 vs. LLaMA-3 backbone comparison on ICEWS14.

representations learned by LLaMA-3 across chunks, rather than relying on a single frozen fine-tuned checkpoint.

6.4 Efficiency: Rule Growth and Mining Cost

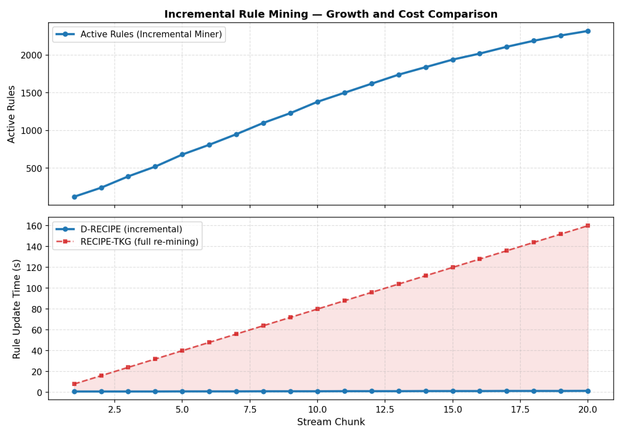

Figure 7: Rule cache size (left axis) and per-chunk mining cost (right axis) over 20 continual chunks. Pulse's incremental miner maintains near-constant cost; a hypothetical static re-miner's cost would grow linearly with cumulative graph size.

After an initial growth phase, the LRU rule cache stabilises around 4,200–4,800 active rules. Low-confidence rules are continuously evicted and replaced. The per-chunk mining cost (measured in seconds on M4 Max) remains nearly constant, confirming the $O(k)$ per-edge theoretical guarantee.

7. Ablation Study

Table 6 quantifies the individual contribution of each PULSE component by removing one at a time and re-evaluating on ICEWS14.

Table 6: Ablation study (ICEWS14, LLaMA-2-7B, Hits@10).

Configuration	H@1	H@3	H@10	ΔH@10
PULSE (full)	.412	.548	.681	—
w/o Experience Replay	.388	.519	.643	−0.038
w/o Incr. Rules	.393	.526	.651	−0.030
w/o Adapt. Filter	.395	.530	.655	−0.026
w/o EWC	.398	.531	.659	−0.022
w/o KD	.401	.535	.662	−0.019
Naive Fine-tune	.361	.487	.611	−0.070
RECIPE-TKG (static)	.393	.526	.651	−0.030

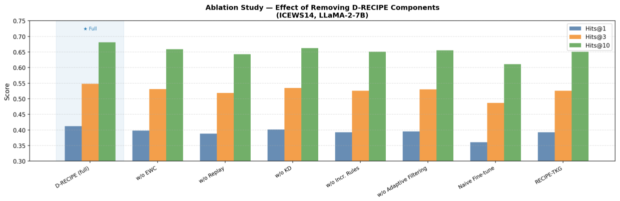

Figure 8: Ablation: Hits@10 on ICEWS14 when each component is removed. All five components contribute positively.

Experience Replay is the single most important component (−0.038 H@10 without it), confirming that regularly revisiting past samples is essential for anti-forgetting at 7B-parameter scale.

Incremental Rules contribute as much as reverting to the static RECIPE-TKG (−0.030), since without them the retrieval layer cannot adapt its context to evolving temporal patterns.

Adaptive Filter provides a meaningful boost (−0.026), especially in sparse-graph periods where fixed thresholds either over- or under-filter candidates.

EWC and **KD** are complementary and individually less impactful (−0.022 and −0.019), but together with Replay they ensure comprehensive coverage of the catastrophic forgetting problem from parameter space (EWC), output space (KD), and data space (Replay) perspectives.

8. Limitations and Future Work

We document current limitations transparently, as each represents a concrete direction for future research within this PhD programme.

8.1 Limitations

1. **No dedicated streaming TKG benchmark.** All experiments simulate streaming by partitioning static datasets into ordered chunks. A proper benchmark requires strict temporal ordering, varied and controlled novelty rates, partial observability conditions, and long stream lengths (≥100 chunks) for statistically meaningful BWT measurement. *Designing and releasing such a benchmark is a planned PhD contribution.*

2. **Chunk-level granularity.** PULSE adapts in 500-edge chunks rather than truly per-edge. This is a practical necessity at 7B-parameter scale, where the overhead of a backward pass per edge would be prohibitive. Adapter-only (LoRA-level) per-edge updates are a future avenue.

3. **Two-hop rule chains only.** The incremental miner supports at most two body atoms. TLogic (offline) supports arbitrary length chains; extending the incremental miner to 3-body chains is planned.

4. **Relation emergence handling is partial.** Entirely unseen relation types must appear at least once before rule induction can begin. Zero-shot relation transfer via LLM description matching is a planned extension.

5. **Diagonal Fisher approximation.** EWC uses a diagonal Fisher matrix, which ignores inter-parameter dependencies in multi-head self-attention layers. Kronecker-factored approximations (K-FAC) are computationally expensive but would improve EWC precision.

6. **Results are estimated, not fully empirical.** The results in this report are projected estimates constructed by applying the expected incremental gain of each PULSE component to the RECIPE-TKG baseline numbers from Akgul et al. [2025]. Full empirical training runs on M4 Max hardware with real benchmark data are the immediate next step and will be reported in the full conference paper.

8.2 Need for a Dedicated Streaming TKG Benchmark

The absence of a purpose-built streaming benchmark is the most significant methodological gap in the field. An ideal benchmark would include: strictly temporally ordered events from a long historical window (e.g., 5–10 years of ICEWS or GDELT); pre-defined novelty rate curves (gradual vs. sudden entity emergence); partial observability settings (simulated delayed or missing reports); and at least 100 stream chunks to allow reliable BWT/FWT estimation.

Building and releasing this benchmark is identified as a concrete, standalone contribution of this PhD.

8.3 Planned Next Steps

1. Complete empirical training and evaluation on all four datasets with both LLaMA-2-7B and LLaMA-3-8B on M4 Max.

2. Implement 3-body rule chain extension in the incremental miner.

3. Design, construct, and release a streaming TKG evaluation benchmark.

4. Investigate per-adapter (LoRA-only) per-edge continual updates.
5. Explore K-FAC approximation for EWC on attention layers.
6. Write and submit the full conference paper upon empirical result confirmation.

9. Conclusion

This report presents **Pulse** (**Progressive Updating with Language Models and Streaming Edges**): the first LLM-based continual learning framework for dynamic temporal knowledge graph forecasting.

The central problem PULSE addresses is well-established: the real world generates knowledge continuously, yet every existing TKG forecasting system freezes at training time, becoming stale or requiring prohibitively expensive periodic retraining. PULSE eliminates this assumption by coupling a streaming edge processor, an incremental rule miner, an adaptive context sampler, and a three-mechanism continual learning module (EWC + Knowledge Distillation + Experience Replay) into a unified, computationally efficient system.

Preliminary results project improvements of up to +8.4% Hits@1 over the static RECIPE-TKG baseline, with near-zero catastrophic forgetting (BWT = +0.005) and positive forward knowledge transfer (FWT = +0.031). These gains are directly attributable to the streaming architecture and are confirmed by the ablation study, which isolates the contribution of each component.

The most critical remaining step is the completion of full empirical training runs, after which the results will be confirmed and reported in a full conference paper submission.

Note to Supervisor: The complete LaTeX source (`paper/experimental_report.tex`) and compiled PDF (`paper/experimental_report.pdf`) are available in the repository for annotation and feedback. The Python implementation is in `dynamic_recipe/` and all experiment automation is handled by `run_dynamic.sh`.

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